the geneticssociety

www.nature.com/hdy

ARTICLE

Mitochondrial structure despite nuclear panmixia: sex-specific dispersal dictates population structure in sperm whales

Reid S. Brennan[1✉], Lynsey A. Wilcox Talbot[2], Anthony Martinez[1], Lance P. Garrison[1], Dan Engelhaupt[3], Nicole L. Vollmer[2,4] and Patricia E. Rosel[2]

This is a U.S. Government work and not under copyright protection in the US; foreign copyright protection may apply 2026

Marine mammals have high potential for dispersal, yet behavioral or environmental constraints can limit gene flow. This is true for the endangered sperm whale, *Physeter macrocephalus*, which has a global distribution and long-distance migrations. While previous studies revealed mitochondrial population structure with weak nuclear structure globally, genomic approaches examining this pattern have been limited. Understanding connectivity is critical for the management of this species due to population declines relative to pre-whaling numbers and increased recent anthropogenic stressors. We investigated connectivity between two regions, the U.S. Gulf of Mexico and the western North Atlantic Ocean, using reduced representation genomic and mitochondrial control region sequencing of 73 sperm whales. Relatedness decreased with geographic distance, likely due to the presence of social groups and familial structure. Nuclear markers showed no population structure ($F_{ST} = 0.001–0.008$), while mitochondrial structure was high ($F_{ST} = 0.36–0.65$), consistent with male-biased dispersal and female philopatry. Female-only analyses showed higher differentiation for mitochondrial but not nuclear markers; male-only analyses revealed no structure. Across all samples, genetic diversity (nuclear: 0.0014; mitochondrial: 0.0017) and effective population size ($N_e = 460$) were low. Given this low diversity and evidence for partitioning of genetic variation, we recommend managers treat these two regions as distinct to preserve existing variation and promote resilience of this species. These results illustrate that despite the increased power of a genomic approach, it is essential to consider the biology of the species at hand and leverage both mitochondrial and nuclear markers to understand the genetic structure of threatened species.

Heredity; https://doi.org/10.1038/s41437-026-00824-x

INTRODUCTION

Marine environments are paradoxical: despite few obvious physical barriers to gene flow and dispersal (Palumbi 1994; Grosberg and Cunningham 2001), many species show population structure and genetic differentiation (Kelly and Palumbi 2010; Sanford and Kelly 2011). While population structure may be expected for organisms with low dispersal potential (Bohonak 1999), this pattern is seen even for species such as cetaceans with extremely high dispersal potential (Ansmann et al. 2012; Vollmer et al. 2021). Understanding these patterns of population structure is critical, particularly for species of conservation or economic concern, where knowledge of population connectivity and structure directly informs management decisions, such as the strategic design of marine reserves (Palumbi 2003; Cros et al. 2017; Bernatchez et al. 2017).

While cetaceans have the ability to swim long distances, they nevertheless show population structure and genetic differentiation. Many species, such as coastal bottlenose dolphins, are philopatric, where individuals remain in their natal area, leading to reduced gene flow between geographic regions (Sellas et al. 2005; Rosel et al. 2009; Speakman et al. 2022). Others, including humpback whales, undergo long-distance migrations to distinct feeding grounds and return to breeding grounds, resulting in genetic structure based on these locations (Baker et al. 2013). Dispersal can also be biased between the sexes, where it is more common to have male-biased dispersal, as has been observed in Indo-Pacific bottlenose dolphins (*Tursiops aduncus*) (Möller and Beheregaray 2004) and Commerson's dolphin (*Cephalorhynchus commersonii*) (Durante et al. 2022). Finally, cetaceans are social animals, and these dynamics can impact dispersal and individual interactions. In some cases, including pilot whales and killer whales, both sexes remain in natal groups, but females tend to mate with individuals from other groups (Amos et al. 1993; Barrett-Lennard 2000; Ford et al. 2011). In other cases, cultural transmission of behavior can impact genetic divergence, as seen between sympatric ecotypes of killer whales (*Orcinus orca*) that are differentiated by prey type (Foote et al. 2016). Thus, despite the potential for high gene flow, genetic differentiation is common in cetaceans and can be driven by multiple mechanisms.

Sperm whales exhibit complex social behaviors that influence their genetic variation. Within social groups, individuals share learned, cultural traits, most notably vocalizations referred to as

[1]Marine Mammal and Turtle Division, Southeast Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Miami, FL, USA. [2]Marine Mammal and Turtle Division, Southeast Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Lafayette, LA, USA. [3]HDR, 701 Independence Circle, Virginia Beach, VA, USA. [4]Cooperative Institute for Marine and Atmospheric Studies, Rosenstiel School for Marine, Atmospheric, and Earth Science, University of Miami, Miami, FL, USA. Associate editor: Bastiaan Star. ✉email: reid.brennan@noaa.gov

Received: 24 April 2025 Revised: 20 January 2026 Accepted: 26 January 2026
Published online: 06 February 2026

codas (Rendell and Whitehead 2003). Further, females are philopatric (Engelhaupt et al. 2009) and maintain long-term familial social groups while males are highly dispersive, show low philopatry, and are relatively solitary after maturity, except during breeding (Whitehead 1993; Richard et al. 1996; Gero et al. 2014; Eguiguren et al. 2023). This sex-biased dispersal pattern is reflected in their genetic structure: mitochondrial markers reveal clear population differentiation across ocean basins (Alexander et al. 2016), while nuclear markers often find weaker or no evidence of structure (Engelhaupt et al. 2009; Mesnick et al. 2011a).

Sperm whales are currently listed as endangered under the U.S. Endangered Species Act and listed globally as Vulnerable on the International Union for Conservation of Nature (Whitehead et al. 2025). Historical global population sizes are estimated to have been around 2 million, but 2022 population sizes were only 845,000 (Whitehead and Shin 2022); this decline has likely been at least partially driven by commercial whaling, for which there has been an international moratorium since 1986. Additionally, there are contemporary threats to the species. For example, whales are exposed to anthropogenic sound from seismic operations related to oil and gas activity as well as high ship traffic (Estabrook et al. 2016) and oil spills (Solsona-Berga et al. 2024); it is likely that these factors have led to local population declines (Farmer et al. 2018). Mitochondrial genetic diversity is low globally (Alexander et al. 2013), and it has been hypothesized that this has been driven by cultural selection and gene-culture coevolution within matrilineal groups (Whitehead 1998). However, more recent work suggests that low genetic variation predates whaling impacts and is likely associated with a species-wide bottleneck approximately 125 thousand years ago (Morin et al. 2018). Regardless of the cause, the complex social dynamics and ongoing threats to sperm whale populations make understanding the patterns of genetic variation essential to effectively manage and conserve this species.

While there has been a global focus on understanding sperm whale population structure, the connectivity between individuals from the U.S. Gulf of Mexico (hereafter "Gulf") and the Atlantic Ocean remains poorly understood. Gulf individuals possess unique codas from Atlantic individuals, suggesting cultural and potentially genetic distinctiveness (Jochens et al. 2008). There is also evidence that codas may be distinct between western and other Gulf individuals, suggestive of genetic differentiation within the Gulf itself (Jochens et al. 2008). Previous work using mitochondrial (mtDNA) and nuclear microsatellite markers has found population structure between Gulf and Atlantic individuals at the mitochondrial, but not the nuclear level (Engelhaupt et al. 2009; Alexander et al. 2013; Lizewski et al. 2025). However, no work has leveraged genomic data using thousands of markers across the genome, which is a key gap as microsatellite markers generally have lower power relative to their genomic counterparts in revealing genetic differentiation (Lah et al. 2016; McCartney-Melstad et al. 2018; Zimmerman et al. 2020; Gallego-García et al. 2021; Forester and Lama 2023; Montana et al. 2024). For example, genomic markers with Mediterranean and Atlantic populations of sperm whales have revealed a more detailed understanding of patterns of divergence than nuclear or mitochondrial markers (Violi et al. 2023). It remains unclear, however, if the lack of nuclear DNA structure between Gulf and Atlantic sperm whales is due to the low power of previous methods or a true lack of structure driven by life history characteristics.

Here, we used a reduced representation approach to characterize the genome-wide genetic variation of 73 sperm whale samples collected from 1995 to 2014 across the northern Gulf and western North Atlantic. By combining these genomic data with additional data from the mitochondrial control region, we reveal the genetic relationships between geographic regions, relatedness between individuals, and the standing levels of genetic variation; factors that are essential for long-term resilience to changing global conditions. These results provide crucial information for developing conservation strategies for this vulnerable species across its U.S. range and demonstrate that nuclear structure is absent between Gulf and Atlantic individuals. Disentangling the population structure of these regions is critical for management purposes, as each area is affected by unique stressors.

METHODS

Skin samples were collected from sperm whales in the northern Gulf and western North Atlantic from 1995 to 2014 and consisted of a mix of stranded individuals ($n = 11$) and remote biopsies (Sinclair et al. 2015) from free swimming animals ($n = 62$), resulting in 73 total samples with 48 females and 25 males (Fig. 1, Supplemental Metadata). Note that all 11 stranded individuals were from the Atlantic coast. DNA was extracted from skin using a Qiagen DNeasy Blood and Tissue kit following the manufacturer's protocol, and four samples with brown-colored elution were further purified with MO BIO's PowerClean DNA Clean-Up Kit (Qiagen) to remove potential PCR inhibitors. DNA quality was determined by gel electrophoresis, and quantity was read on a fluorometer (GE HealthCare Hoefer DyNA Quant 200 Fluorometer). A subset of the biopsy samples ($n = 40$) were previously analyzed for mtDNA and microsatellites as well as sex (Engelhaupt et al. 2009) and sex of the remaining biopsies were genetically determined by PCR amplification of ZFX and SRY gene fragments as described by Rosel (2003), except 25 ng of DNA and 0.75 U of *Taq* DNA Polymerase (Invitrogen) were used in each reaction. The primers for SRY were 5′-CATTGTGTGGTCTCGT-GATC-3′ and 5′-ACCGGCTTTCCATTCGTGAACG-3′, and for ZFX were 5′-ATAGGTCTGCAGACTCTTCTA-3′ and 5′-AGAATATGGCGACTTAGAACG-3′. The amplified fragments were visualized by gel electrophoresis on a 2.5% agarose gel in 1× SB (sodium borate) buffer. The sex of stranded individuals was determined in the field through visual inspection. Samples were separated into four putative geographic regions based on their sampling location and hypothesized population structure and separation: Atlantic, Dry Tortugas, Northern Gulf (N. Gulf), and Western Gulf (W. Gulf) (Fig. 1).

Genomic data were collected using a reduced representation genomic approach, Nextera-tagmented reductively-amplified DNA (nextRAD) genotyping-by-sequencing from SNPsaurus (LLC). This method uses selective PCR primers to repeatedly target the same genomic regions across all samples. Genomic DNA was randomly fragmented with the Nextera reagent (Illumina) and then amplified with a custom primer targeting the 10-base pair (bp) sequence GTGTAGAGCC. Resulting libraries were sequenced on one lane of an Illumina HiSeq 4000 with 150 bp single-end reads. See Russello et al. (2015) for full details of nextRAD.

Raw reads were quality checked with fastQC v0.11.9, trimmed for quality and adapter contamination using *fastp* v0.23.4 (Chen et al. 2018), and aligned to the *Physeter macrocephalus* reference genome (Fan et al. 2019) using bwa-mem2 v2.2.1 (Vasimuddin et al. 2019). Variants were called with freebayes v1.3.6 (Garrison and Marth 2012) and stringently filtered using VCFtools v0.1.16 (Danecek et al. 2011) to keep only sites that were: bi-allelic, minor allele frequency greater than 0.05, less than 30% missing

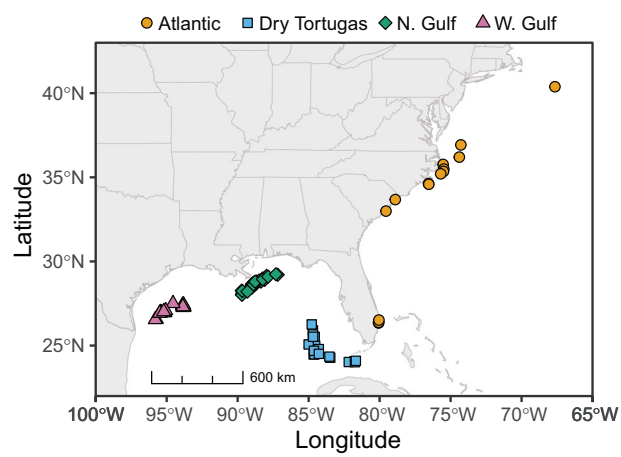

Fig. 1 Collection sites for all samples. Each point is an individual sample, and points are colored by geographic region corresponding to the a priori hypothesized population structure.

data, and a minimum mean depth of 10 and a maximum of 53. Finally, potential paralogs were removed using HDplot, removing sites with heterozygosity greater than 0.6 and read-ratio deviations greater than 6 (McKinney et al. 2017). This resulted in 4441 single-nucleotide polymorphisms (SNPs). See accompanying scripts on Zenodo for full filtering details.

Related individuals were identified using ngsRelateV2 (Hanghøj et al. 2019) using both the KING-robust kinship estimator (Waples et al. 2019) and the coefficient of coancestry Θ (Jacquard 1974). Individuals were considered related if they were at least 3rd-degree relatives, with a kinship coefficient for KING greater than 0.0442 (Manichaikul et al. 2010) and Θ greater than 0.0625 (Jacquard 1972). We also took a more lenient approach, considering individuals related if they had either a kinship coefficient greater than 0.05 or Θ greater than 0.05. Individuals with Θ of ~0.25 were considered full siblings and 0.125 half-siblings. To distinguish full siblings from parent-offspring pairs, we used the method described by Waples et al. (2019), leveraging the R0 statistic, which is the proportion of unique homozygous loci to shared heterozygous loci. Here, we expect R0 to be zero for parent-offspring pairs because they cannot possess unique homozygous loci, but greater than zero for siblings. We tested for a relationship between relatedness (Θ) and ocean distance between samples using a zero-inflated beta regression using the GAMLSS framework (Rigby and Stasinopoulos 2005) in R v4.4.1 (R Core Team 2024). This approach was chosen as Θ is bounded between zero and one and contains zeros. Pairwise ocean distances between all individuals were calculated by finding the least-cost path using the marmap package (Pante and Simon-Bouhet 2013), limiting minimum depth to 0.3 meters to avoid crossing land; paths were visually checked to ensure land was not crossed. While some individuals were stranded, these location estimations represent an approximate location of the individuals prior to stranding, though the distance any individual drifted prior to stranding cannot be known.

To understand the potential genetic structure among geographic regions, we first conducted a principal component analysis (PCA). For all analyses of population structure, individuals were removed such that no related samples remained in the dataset, leaving 61 unrelated individuals (42 females, 19 males). Using PLINK v1.90, variants were thinned for linkage disequilibrium (LD) with the option –indep 50 5 2, and a PCA was run on the resulting 3977 SNPs. Ancestries of each individual were estimated using Admixture v1.3.0 (Alexander et al. 2009), testing one to eight populations, and the most likely number of populations was determined based on the lowest cross-validation error. We repeated these same analyses with only the 42 female individuals. Pairwise F_{ST} among geographic regions was calculated using Weir and Cockerham's method (Weir and Cockerham 1984) implemented in snpR v1.2.11 with 500 permutation replicates to calculate significance (Hemstrom and Jones 2023).

Population structure was also tested using discriminant analysis of principal components (DAPC). One of the most important parameters in running DAPC is the number of principal components (PC) axes to include as the predictor. Thia (2023) recently showed that the number of PCs should be equal to $k-1$, where k is the number of effective populations in the dataset (Thia 2023). We applied this principle as well as the cross-validation and *a*-score approach described by the authors of the adegenet package (Jombart 2008). The number of populations was estimated using K-means clustering (*find.clusters()*) where the most likely k was identified as the lowest Bayesian information criterion (BIC) score, and a system with population structure should give an "elbow" where BIC values decrease until the most likely k and then plateau or increase. This pattern should be robust to changes in the number of PCs used when structure is present. Following this, we tested the hypotheses that the number of populations is equal to four, the number of geographic locations is equal to two, and the split between Gulf and Atlantic individuals. We chose to include the N. Gulf and W. Gulf as Gulf samples and the Atlantic and Dry Tortuga as Atlantic individuals when considering two populations. This was based on the mitochondrial structure results (below) as well as previous work that suggests Dry Tortuga individuals are more similar to Atlantic than Gulf individuals (Lizewski et al. 2025). We assessed the fit of each of these models using 100 replicates of training-testing partitioning as recommended by Thia (2023). For each replicate, we randomly assigned 70% of the samples to the training partition and built a DAPC model. We then predicted the population source of the other 30% of individuals (testing partition) given this model. Assignment accuracy was calculated as the proportion of testing individuals assigned back to their population of origin. With a two-population model, random assignment from an uninformative model would result in 50% accuracy, while a four-population model would result in 25% accuracy.

Across all calculations of genetic diversity and inbreeding, potential siblings were retained in the dataset to avoid inducing bias (Waples and Anderson 2017). All values were calculated by the putative geographic population as well as across all individuals. Pairwise genetic diversity (π) was estimated in 50 kb windows across assembled chromosomes (and not unassembled scaffolds) using *pixy* v1.2.10 (Korunes and Samuk 2021). Both variant and invariant sites were called with freebayes v1.3.6 and filtered by depth and missingness in the same manner as described above. Including invariant sites is critical to avoid bias in estimates of π. Confidence intervals were determined using 10,000 bootstrap replicates, and results were robust when window sizes were decreased to 10 kb. F_{IS} and expected and observed heterozygosity were calculated using Stacks v2.65 (Rochette et al. 2019). Contemporary effective population size was estimated using *currentNe* v1.0 (Santiago et al. 2024), using only loci located on assembled chromosomes and both with and without specifying the average number of siblings an individual has in the sample (0.11), as well as with NeEstimator v2 (Do et al. 2014) implemented via dartR v2.9.8 (Mijangos et al. 2022).

To quantify mitochondrial genetic diversity and population structure, we sequenced the mtDNA control region for all 73 samples. The control region and flanking transfer RNA (tRNA) (1167 bp) was PCR amplified using primers L15928 (Vollmer et al. 2011) and a modified version of H00651 (5′-CAGAAGGCTAGGACCAAACCT-3′, modified from Kocher 1989) and Phusion™ High-Fidelity DNA Polymerase (Thermo Scientific). Each PCR was performed in a 25 µL reaction with 1× Phusion™ HF Buffer, 0.5 U Phusion™ Polymerase, 0.5 µM of each primer, 200 µM dNTPs, and 25 ng of DNA. The thermal cycling profile consisted of an initial denaturation step of 98 °C for 30 s followed by 30 cycles of 98 °C for 10 s, 63 °C for 20 s, and 72 °C for 20 s with a final extension at 72 °C for 5 min. Amplified products were purified via extraction from low-melting-point agarose gels followed by digestion using agarase (Sigma-Aldrich), and then sequenced in both directions using a BigDye Terminator v3.1 Cycle Sequencing Kit (Thermo Scientific) on an ABI 3130 Genetic Analyzer. The forward and reverse sequences were independently edited, and then consensus sequences were assembled using Sequencher v5.3 (GeneCodes) or Geneious v9.0.2 (https://www.geneious.com). Sequences were aligned using Se-Al 2.0a11 (Rambaut 2002) or Geneious v9.0.2, and flanking regions were truncated to include only the control region sequence of 955 bp. Nucleotide diversity was calculated for all samples together and as separate populations using Pegas v1.3 (Paradis 2010). Hierfstat v0.5.11 was used to calculate pairwise F_{ST} between populations (Goudet 2005), and permutations shuffling population labels were used to determine if observed F_{ST} values were greater than expected by chance. Note that Hierfstat was used for mitochondrial data as it can handle haploid markers, while snpR cannot. Statistical parsimony haplotype networks (i.e., TCS for Templeton, Crandall, and Sing 1992) were calculated using PopART (Leigh et al. 2015). Finally, we ran an Analysis of Molecular Variance (AMOVA) using ade4 v1.7.22 (Dray and Dufour 2007) implemented in poppr v2.9.6 (Kamvar et al. 2014).

RESULTS

Relatedness

There were four pairs of first-degree full siblings (seven females, one male; Fig. S1), eight pairs of second-degree relatives (nine females, seven males), and four pairs of third-degree relatives (Fig. 2; seven females, one male); the number of third-degree relatives should be treated cautiously as these relationships can be difficult to identify (Manichaikul et al. 2010). Full siblings were sampled within a geographic region, where 3 pairs were sampled within the W. Gulf and one within the N. Gulf (Figs. 2 and S1). Overall, closely related individuals were more likely to be sampled at closer geographic distances, with relatedness (Θ) decreasing by 0.036 per 1000 km ($p = 0.003$), likely reflecting the familial structure of sperm whales (see "Discussion" section).

Nuclear DNA structure and diversity

There was no evidence for population structure when looking at the nuclear nextRAD markers. Principal component analysis showed panmixia and no clustering by population (Figs. 3A and S2). Similarly, F_{ST} was nearly zero between all populations ($F_{ST} = 0.001–0.008$; Fig. 3B, above diagonal) and not significantly different from the null hypothesis of panmixia (Fig. 3B, below diagonal). The most likely number of populations was

one, as indicated by the cross-validation error from Admixture (Fig. S3). Finally, DAPC revealed no power to discriminate among populations using either a two- or four-population model and regardless of the approach for the number of PCs included (Figs. 3C and S4). The training-testing partitioning was no better than random in assigning individuals to a population, indicating a lack of discriminatory genetic differences among populations (Fig. 3C). This result was robust to the number of PCs included in the DAPC model (Fig. S4). Further, K-means clustering showed no "elbow" indicative of population structure (Fig. S5) and when adding PCs to the DAPC model the slope of the curve switched from negative (low PCs) to positive (high PCs), mirroring simulation results from Thia (2023) showing the same pattern in panmictic populations; systems with population structure are generally robust to changes in the number of PCs included in the model. Note that the lack of population structure is consistent when analyzing the unrelated females without any males (Figs. S6 and S7).

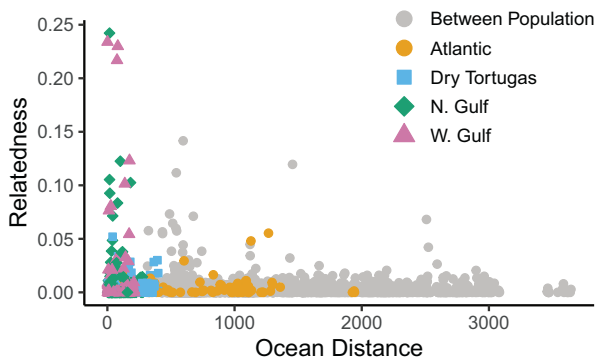

Fig. 2 Pairwise relatedness between all individuals colored by the population origin of each individual. Relatedness is Θ, and distance is the shortest ocean distance in km, avoiding land between two individuals. There was a negative relationship between ocean distance and relatedness, where individuals collected at close ocean distances were more likely to be closely related than those at farther distances ($p = 0.003$).

For the nuclear markers, across all individuals, pairwise genetic diversity (Tajima's π) was 0.00136 (95% confidence interval: 0.00133, 0.00140; Fig. 4A). As there was no evidence of population structure, all individuals should be treated as a single population for these calculations. For completeness, we also provide location-level estimates: Atlantic: $\pi = 0.00137$ [0.00133, 0.00141]; Dry Tortugas: $\pi = 0.00133$ [0.00130, 0.00137]; N. Gulf: $\pi = 0.00140$ [0.00136, 0.00143]; W, Gulf: $\pi = 0.00132$ [0.00128, 0.00136]. Observed heterozygosity was slightly lower than expected heterozygosity across all individuals ($H_{obs} = 0.263$ [0.261, 0.265]; $H_{exp} = 0.283$ [0.280, 0.284]; Fig. 4B, C). Note there was no temporal effect in the data as heterozygosity was consistent across the sampling period with early and late samples, delineated at June 16, 2005, both having H_{obs} of 0.263. Estimates were largely consistent across location, with the exception of Atlantic, which was lower than the others ($H_{obs} =$ 0.252 [0.250, 0.255]; $H_{exp} = 0.270$ [0.267,0.272]). The inbreeding coefficient (F_{IS}) was 0.073 and did not overlap with zero [0.036, 0.111], indicating the presence of inbreeding. Again, there were a few differences across locations except that Atlantic ($F_{IS} = 0.0799$ [0.067, 0.092]) was higher than the W. Gulf ($F_{IS} = 0.052$ [0.039, 0.065]). Contemporary effective population size estimated across all geographic locations using CurrentNe was 459.54 (90% CI: 324.57, 650.64) without specifying the number of siblings in the sample and 708.52 (90% CI: 469.47, 1069.29) when specifying the average number of siblings in the sample. NeEstimator provided similar results (516.6) but with infinite confidence intervals, likely due to the sampling error correction (Do et al. 2014). When subset to just the Gulf, CurrentNe estimated effective population size as 184.03 (90% CI: 123.44, 274.35) and when accounting for siblings, 333.27 (90% CI: 201.59, 550.95). Conversely, Atlantic individuals (Atlantic + Dry Tortugas) had an effective population size of 646.94 (90% CI: 289.01, 1448.16); no full siblings were found in these populations, so the model with the sibling parameter was not run.

Mitochondrial DNA structure and diversity

Similar to the nuclear markers, there was low overall mitochondrial nucleotide diversity (0.0018 [0.0014, 0.0019]) and no significant

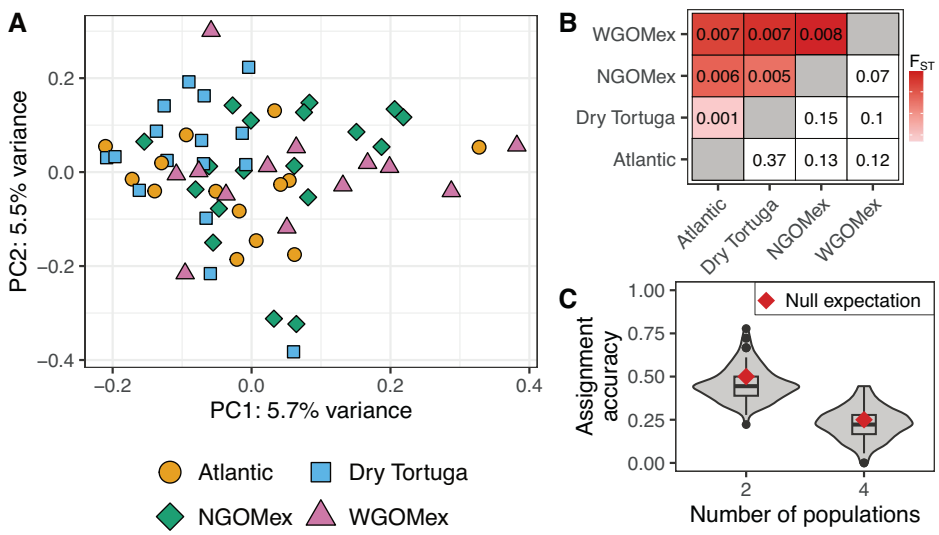

Fig. 3 Nuclear panmixia across populations. A Principal components analysis of 3977 LD-thinned SNPs showing a lack of clustering by geographic location. Geographic locations are represented by shape and color. **B** Pairwise F_{ST} estimates between locations. Above diagonal and colors indicate F_{ST} values, while below the diagonal represent p-values from 500 permutations testing panmixia. **C** Assignment accuracy from DAPC following 100 replicates of training-testing partitioning, assuming both two (Atlantic: Atlantic + Dry Tortugas; Gulf: N. Gulf + W. Gulf) and four populations. If population assignments were completely random, we expect 50% accuracy for the two-population model and 25% accuracy for the four-population model. The boxplot and violin plot show the distribution of mean accuracy for the 100 replicate runs, and the red diamond is the null expectation given random assignments.

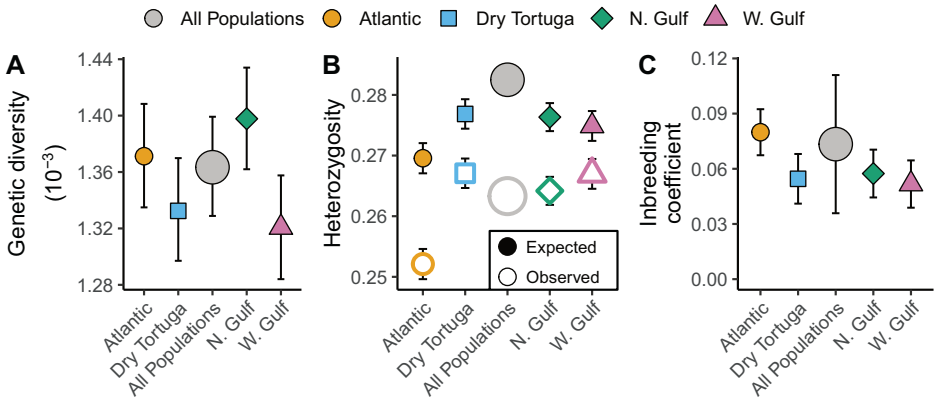

Fig. 4 Measures of genetic diversity for nuclear markers. A Pairwise genetic diversity, (**B**) observed and expected heterozygosity, and (**C**) inbreeding coefficients (F_{IS}). Values were estimated considering all individuals as a single population ("All populations"), as suggested by population structure results, as well as by geographic location. For heterozygosity (**B**), filled points indicate expected heterozygosity, and empty points are observed heterozygosity. For all plots, the error bars indicate the 95% confidence interval, where non-overlapping intervals are significantly different; see the text for these values.

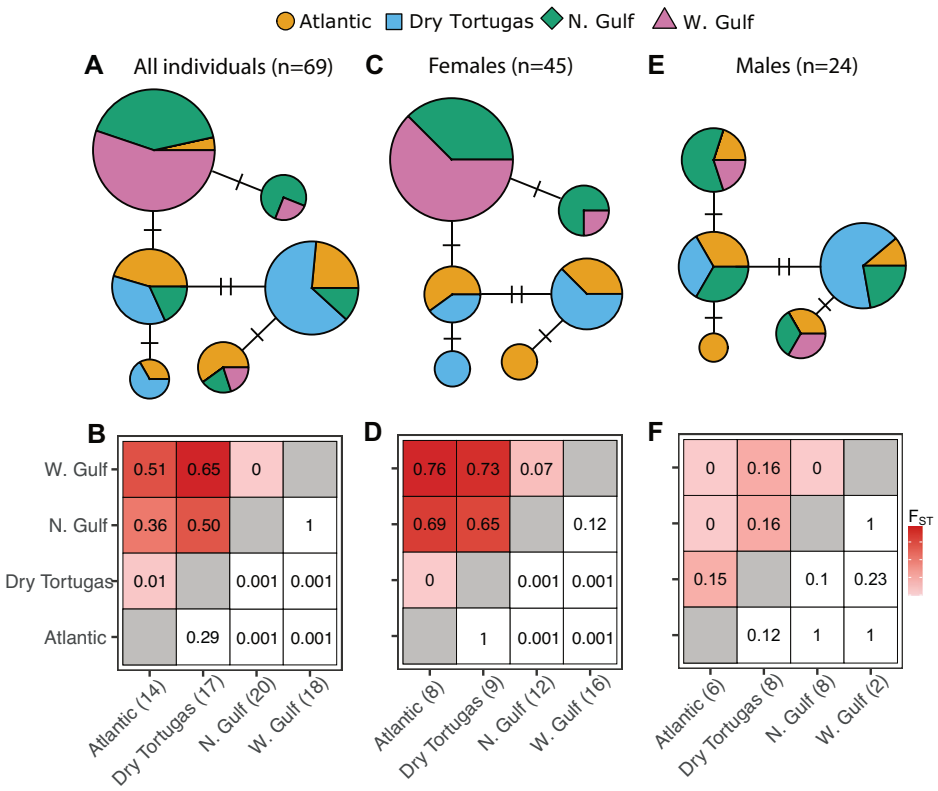

Fig. 5 mtDNA population structure across all individuals and subsets by sex. Haplotype networks (**A**, **C**, **E**) were calculated via the TCS method, where the size of the circle indicates the number of individuals possessing each haplotype and color indicates the population origin. Hash marks denote the number of mutations between haplotypes. Four individuals were heteroplasmic and removed from the analysis. Measures of pairwise F_{ST} between populations (**B**, **D**, **F**), where the above diagonal red triangle indicates F_{ST} values and the lower diagonal includes *p*-values from 1000 permutations testing the null hypothesis of panmixia. Numbers after the population name on the x-axis indicate the number of individuals per population.

difference in levels of diversity among locations (Fig. S8). However, in contrast to the nuclear DNA, there was significant evidence for population structure (Fig. 5). The haplotype network with all individuals (Fig. 5A) showed significant population structure with 38.7% of the genetic variation occurring between locations and 61.3% occurring within locations (AMOVA, $p = 0.001$). F_{ST} between Atlantic and Dry Tortugas as well as between N. Gulf and W. Gulf were low and not significantly different from the null hypothesis of panmixia (Fig. 5B; Atlantic vs. Dry Tortugas: $F_{ST} = 0.01$, $p = 0.29$; N. Gulf vs. W. Gulf: $F_{ST} = 0.00$, $p = 1.00$). In contrast, F_{ST} was high between Atlantic or Dry Tortugas versus N. Gulf or W. Gulf ($F_{ST} = 0.36$–0.65, $p = 0.001$; Fig. 5B). Population structure was much stronger when considering only females (Fig. 5C, D). No haplotypes were shared between N. and W. Gulf compared to Dry Tortugas and Atlantic, and accordingly, F_{ST} was higher than all individual analyses (Fig. 5D; $F_{ST} = 0.65$–0.76; $p = 0.001$). However, there was still no structure within the Gulf ($F_{ST} = 0.07$, $p = 0.12$) or Atlantic ($F_{ST} = 0$, $p = 1$). In contrast, the male analysis showed no evidence for

population structure for any comparisons (Fig. 5E, F; $F_{ST} = 0–0.16$, $p > 0.1$). However, sample sizes were small after subsetting, especially for the male analysis, where W. Gulf had only two individuals, making these results less robust than the full analysis (see Fig. 5 for sample numbers). Together, these results indicate significant population structure for mtDNA between the Gulf versus Dry Tortugas and Atlantic, particularly for females.

DISCUSSION

Here, we demonstrate that the sperm whale, *Physeter macrocephalus*, shows significant population structure in its mitochondrial DNA despite panmixia in its nuclear genome across the Northern Gulf and western North Atlantic Ocean. This result is likely driven by the unique life history of sperm whales; there is a strong family structure where females typically associate with their familial group for their entire lives, and dispersal is predominantly male-mediated (Best 1979). These results stress the importance of considering the life history of species and marker choice when assessing population structure, particularly for species of conservation concern such as sperm whales. Finally, our findings indicate that sperm whales in the Gulf are distinct from those in the Atlantic and should be considered separate units for management purposes.

Despite genotyping 4441 nuclear loci across 73 samples, we found no evidence of population structure in the nuclear genome in this species between the Gulf and western North Atlantic. We evaluated whether the general lack of population structure for the nuclear genome in previous studies across various regions was due to a lack of power of microsatellite markers to detect fine-scale structure (Lyrholm et al. 1999; Engelhaupt et al. 2009; Mesnick et al. 2011b; Alexander et al. 2016; Lizewski et al. 2025). Microsatellite loci are typically well suited to detect population structure due to their high mutation rate and allelic diversity and relatively low genotyping costs (Guichoux et al.2011). However, microsatellites are usually limited to 10 s of loci (Hodel et al.2016) and under most scenarios, especially weak population differentiation, increasing the number of markers using genomic approaches can increase the power to detect this structure (Jeffries et al. 2016; Gallego-García et al. 2021; Poelstra 2022). Similarly, increasing the number of loci can improve the ability to identify related individuals (Lemopoulos et al. 2019) and allows for accurate recovery of population structure and genetic diversity using fewer individual samples per population (Nazareno et al. 2017; Gallego-García et al. 2021). This final point is critical for sperm whales and other marine mammals, where obtaining high sample numbers is particularly challenging and expensive. Despite the thousands of loci used in this study, there was no evidence of population structure within the nuclear genome. However, it is possible that whole-genome sequencing would identify more subtle signals of differentiation than we were able to identify in this study.

Previous work on sperm whales has shown limited mtDNA structure across small spatial scales but differentiation across ocean basins (Morin et al. 2018), which is consistent with the results here. Further, both Lizewski et al. (2025) and Engelhaupt et al. (2009) quantified mtDNA structure between Gulf and Atlantic sperm whales and found similar population structure, including the separation of Dry Tortuga samples from the Gulf. The number of haplotypes identified between these studies was also consistent, where Lizewski also found only six haplotypes, sampling 377 individuals. While the low number of haplotypes and overall mitochondrial diversity in this study may inflate the F_{ST} estimates (Charlesworth 1998), the presence of structure via F_{ST} was robust to permutations and under an AMOVA framework, suggesting that this result is not due to low diversity alone. Full mitochondrial genomes would be better able to resolve population structure than the control region fragment used here (Morin et al. 2018), and increasing the data to include mitochondrial genomes would likely increase both the number of haplotypes present and the evidence for population structure.

The presence of mtDNA but not nuclear population structure is likely driven by the distinct sex-specific life history and dispersal patterns of sperm whales; male-mediated dispersal and connectivity generate nuclear DNA panmixia but female philopatry leads to mtDNA structure. Indeed, when separating the data into male and female groups, the nuclear markers showed no structure for either sex (Figs. 3, S3–S7). In contrast, mtDNA structure increased for females but was not present for males (Fig. 5). This strong signal likely represents local social groups of females where individuals are genetically and culturally similar within regions (Jochens et al. 2008). Sperm whales have a multi-generational social structure and associations that consist of adult females and juveniles of both sexes. While males are found within these groups as juveniles, after reaching adolescence, they disperse and become increasingly solitary with large ranges that extend to higher latitudes (Whitehead 2018); males have been tracked moving between the Gulf and the Atlantic (Jochens et al. 2008). Males return to warmer waters for breeding and associate with females and regions for only a short period of the year (Eguiguren et al. 2023). Furthermore, male philopatry seems limited as individuals are only rarely observed multiple times in the same region (Whitehead 1993; Gero et al. 2014; Girardet et al. 2022). Thus, the broad dispersal and general lack of philopatry of males would explain the lack of nuclear population structure in this species.

In contrast to males, satellite tracking of sperm whales in the Gulf over weeks to almost two years has shown that females display philopatry and do not leave the region (Ortega-Ortiz et al. 2012). Previous work has found mtDNA structure between Gulf and Atlantic populations (Engelhaupt et al. 2009; Lizewski et al. 2025) as well as globally (Alexander et al. 2016; Morin et al. 2018), but very limited differentiation in the nuclear genome using microsatellite markers (Engelhaupt et al. 2009; Alexander et al. 2016). Similar patterns of mitochondrial but weak or missing nuclear population structure have been found in other species such as humpback whales (Baker et al. 2013), killer whales (Pilot et al. 2010), beluga whales (O'Corry-Crowe et al. 2018), and Port Jackson sharks (Day et al. 2019).

There was significant evidence that related individuals were more likely to be found at close geographic distances (Fig. 2), consistent with the life history of sperm whales. We found a mix of males and females within the 16 pairs of closely related individuals (22 females, 8 males), and these individuals were generally within the Gulf region. Thirteen of the 16 pairs shared a mitochondrial haplotype, with all first-degree relatives sharing the same haplotype. This suggests that local social units were sampled at each site and may be persistent in those sites. In this study, it is possible that the highly related individuals found here were members of the same group sampled at different times, which is particularly likely for the first-degree relatives. Previous work in the Caribbean has found that 82.5% of individuals within a social unit have a first-degree relative in that unit, but has not found first-degree relatives between social groups (Konrad et al. 2018). Other work in different regions has found a high proportion of related individuals within groups, including in Mauritius (Sarano et al. 2021) and the Azores (Pinela et al. 2009). Similarly, within social units, relatives are more likely to associate with calves (i.e., "babysit") than non-relatives (Konrad et al. 2019). Our findings support this conclusion as the first-degree relatives in this study were likely sampled from the same social groups with samples coming from the same day 0.5 km apart (Pmac097 and Pmac098; $\Theta = 0.23$), six days and 16 km apart (Pmac031 and Pmac125; $\Theta = 0.24$), three days and 73 km apart (Pmac100 and Pmac102; $\Theta = 0.22$), and 12 days and 89 km apart (Pmac093 and Pmac102; $\Theta = 0.23$). Finally, the "isolation-by-distance" pattern observed here, where the probability of relatedness decays with geographic distance, likely reflects the uniqueness of the Gulf region in group

social cohesion relative to the Atlantic. That is, the social groups appear to be relatively consistent within the Gulf and likely explain the patterns of both relatedness and shared mitochondrial haplotypes within this region.

Genetic diversity was low for both nuclear and mitochondrial markers. Nuclear genetic diversity (π) was 0.00136 (0.00133, 0.00140), which is on the lower end for mammals (Buffalo 2021). Similarly, mtDNA nucleotide diversity was 0.0018 [0.0014, 0.0019], which is slightly higher than the 0.001 estimate from Morin et al. (2018) based on 11 whole mitochondrial genomes and similar to the 0.0016 in the Gulf of California found by Cruz-Villagran (2024) using the mitochondrial control region. Only 6 haplotypes were found, which is somewhat lower than previous studies and may be related to the lower number of samples here. This low mitochondrial diversity is likely driven by a bottleneck that occurred about 125 thousand years ago and an expansion over the past 20–40 thousand years (Morin et al. 2018). Petit-Marty et al. (2021) showed that threatened mammal species have diversity levels of 0.0044–0.0102 at cytochrome oxidase, which is less variable than the control region used in this study. Our estimate of 0.0018 is lower than the lower end of this threatened species range. Further, positive inbreeding coefficients (F_{IS}; Fig. 4D) and relatively low effective population size ($N_e = 459.54$ [324.57, 650.64]) suggest that some degree of inbreeding is present. As might be expected given the more isolated Gulf region or potential impacts from the *Deepwater Horizon* oil spill, N_e was lower for Gulf individuals (184.03 [123.44, 274.35]) as compared to Atlantic (646.94 [289.01, 1448.16]). These values are potentially driven by the strong social structure of females, reducing the effective number of breeders and decreasing N_e. While N_e values nearly satisfy the "50/500" rule, where an N_e of 50 is required to prevent inbreeding and 500 to maintain evolutionary potential and long-term viability (Franklin 1980), more recent work shows that an N_e of 1000 (Frankham et al. 2014) or even 5000 (Lande 1995) is required for long-term viability of a population. Therefore, N_e is below what is considered resilient over the long term for this species, particularly in the Gulf. Given the substantial reductions in population numbers to about one-third of pre-whaling sizes, it is key to continue to prioritize population recovery and protection to prevent further decreases in effective population size and long-term viability.

Finally, the mtDNA population structure and low genetic diversity presented here support the current delineation of the species into separate Northern Gulf and western North Atlantic stocks (Hayes et al., 2024). While there is gene flow across the nuclear genome, the limited mitochondrial gene flow indicates that sperm whales from the Gulf are a distinct mitochondrial lineage from the western North Atlantic, including the Dry Tortugas, and should be treated as separate groups for management purposes. However, additional samples are needed to more accurately determine the geographic boundary between these populations and to understand the wider connectivity among animals throughout the Gulf of Mexico and the Caribbean. Given the low mitochondrial diversity in this species, coupled with the low effective population size in the Gulf, preserving the current diversity across both the Gulf and Atlantic is likely key to maintaining the resilience of this species over the long term.

DATA AVAILABILITY

Raw genomic data can be found on NCBI under bioproject PRJNA1189584. Code to run all analyses and metadata for samples, including NCBI accessions for mitochondrial data, can be found on Zenodo: https://doi.org/10.5281/zenodo.14715105.

REFERENCES

Alexander DH, Novembre J, Lange K (2009) Fast model-based estimation of ancestry in unrelated individuals. Genome Res 19:1655–1664

Alexander A, Steel D, Hoekzema K, Mesnick SL, Engelhaupt D, Kerr I et al. (2016) What influences the worldwide genetic structure of sperm whales (*Physeter macrocephalus*)?. Mol Ecol 25:2754–2772

Alexander A, Steel D, Slikas B, Hoekzema K, Carraher C, Parks M et al. (2013) Low Diversity in the mitogenome of sperm whales revealed by next-generation sequencing. Genome Biol Evol 5:113–129

Amos B, Schlötterer C, Tautz D (1993) Social structure of pilot whales revealed by analytical DNA profiling. Science 260:670–672

Ansmann IC, Parra GJ, Lanyon JM, Seddon JM (2012) Fine-scale genetic population structure in a mobile marine mammal: inshore bottlenose dolphins in Moreton Bay, Australia. Mol Ecol 21:4472–4485

Baker CS, Steel D, Calambokidis J, Falcone E, González-Peral U, Barlow J et al. (2013) Strong maternal fidelity and natal philopatry shape genetic structure in North Pacific humpback whales. Mar Ecol Prog Ser 494:291–306

Barrett-Lennard LG (2000) Population structure and mating patterns of Killer Whales (*Orcinus orca*) as revealed by DNA analysis. Dissertation, University of British Columbia

Bernatchez L, Wellenreuther M, Araneda C, Ashton DT, Barth JMI, Beacham TD et al. (2017) Harnessing the power of genomics to secure the future of seafood. Trends Ecol Evol 32:665–680

Best PB (1979) Social organization in sperm whales, *Physeter macrocephalus*. In: Winn HE, Olla BL (eds) Behavior of marine animals. Springer US, Boston, MA, pp 227–289

Bohonak AJ (1999) Dispersal, gene flow, and population structure. Q Rev Biol 74:21–45

Buffalo V (2021) Quantifying the relationship between genetic diversity and population size suggests natural selection cannot explain Lewontin's paradox. eLife 10: e67509

Charlesworth B (1998) Measures of divergence between populations and the effect of forces that reduce variability. Mol Biol Evol 15:538–543

Chen S, Zhou Y, Chen Y, Gu J (2018) fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics 34:i884–i890

Cros A, Toonen RJ, Donahue MJ, Karl SA (2017) Connecting Palau's marine protected areas: a population genetic approach to conservation. Coral Reefs 36:735–748

Cruz-Villagrán H, Blanco-Jarvio A, Silva-Segundo CA, Ramos-Rodríguez JA, Flores-Ramírez S, Galván-Tirado C et al. (2024) Mitochondrial DNA diversity of sperm whales (*Physeter macrocephalus*) in the Gulf of California under a changing environment. Reg Stud Mar Sci 73:103504

Danecek P, Auton A, Abecasis G, Albers CA, Banks E, DePristo MA et al. (2011) The variant call format and VCFtools. Bioinformatics 27:2156–2158

Day J, Clark JA, Williamson JE, Brown C, Gillings M (2019) Population genetic analyses reveal female reproductive philopatry in the oviparous Port Jackson shark. Mar Freshw Res 70:986–994

Do C, Waples RS, Peel D, Macbeth GM, Tillett BJ, Ovenden JR (2014) NeEstimator v2: re-implementation of software for the estimation of contemporary effective population size (N_e) from genetic data. Mol Ecol Resour 14:209–214

Dray S, Dufour A-B (2007) The ade4 package: implementing the duality diagram for ecologists. J Stat Softw 22:1–20

Durante CA, Loizaga R, McCracken GR, Crespo EA, Ruzzante DE (2022) Commerson's dolphin population structure: evidence for female phylopatry and male dispersal. Sci Rep 12: 22219.

Eguiguren A, Konrad Clarke CM, Cantor M (2023) Sperm whale reproductive strategies: current knowledge and future directions. In: Würsig B, Orbach DN (eds) Sex in Cetaceans: Morphology, Behavior, and the Evolution of Sexual Strategies. Springer International Publishing: Cham, pp 443–467

Engelhaupt D, Rus Hoelzel A, Nicholson C, Frantzis A, Mesnick S, Gero S et al. (2009) Female philopatry in coastal basins and male dispersion across the North Atlantic in a highly mobile marine species, the sperm whale (*Physeter macrocephalus*). Mol Ecol 18:4193–4205

Estabrook B, Ponirakis D, Clark C, Rice A (2016) Widespread spatial and temporal extent of anthropogenic noise across the northeastern Gulf of Mexico shelf ecosystem. Endanger Species Res 30:267–282

Fan G, Zhang Y, Liu X, Wang J, Sun Z, Sun S et al. (2019) The first chromosome-level genome for a marine mammal as a resource to study ecology and evolution. Mol Ecol Resour 19:944–956

Farmer NA, Baker K, Zeddies DG, Denes SL, Noren DP, Garrison LP et al. (2018) Population consequences of disturbance by offshore oil and gas activity for endangered sperm whales (*Physeter macrocephalus*). Biol Conserv 227:189–204

Foote AD, Vijay N, Ávila-Arcos MC, Baird RW, Durban JW, Fumagalli M et al. (2016) Genome-culture coevolution promotes rapid divergence of killer whale ecotypes. Nat Commun 7: 11693

Ford MJ, Hanson MB, Hempelmann JA, Ayres KL, Emmons CK, Schorr GS et al. (2011) Inferred paternity and male reproductive success in a killer whale (*Orcinus orca*) Population. J Hered 102:537–553

Forester BR, Lama TM (2023) The role of genomics in the future of Endangered Species Act decision-making. Codex Endanger Species Act 2:2

Frankham R, Bradshaw CJA, Brook BW (2014) Genetics in conservation management: Revised recommendations for the 50/500 rules, Red List criteria and population viability analyses. Biol Conserv 170:56–63

Franklin I (1980) Evolutionary change in small populations. In: Conservation biology - an evolutionary-ecological perspective. 135–149 (Sinauer Associates, Sunderland, MA)

Gallego-García N, Caballero S, Shaffer HB (2021) Are genomic updates of well-studied species worth the investment for conservation? A case study of the critically endangered Magdalena River Turtle. J Hered 112:575–589

Garrison E, Marth G (2012) Haplotype-based variant detection from short-read sequencing. preprint arXiv:1207.3907, https://arxiv.org/abs/1207.3907

Gero S, Milligan M, Rinaldi C, Francis P, Gordon J, Carlson C et al. (2014) Behavior and social structure of the sperm whales of Dominica, West Indies. Mar Mammal Sci 30:905–922

Girardet J, Sarano F, Richard G, Tixier P, Guinet C, Alexander A et al. (2022) Long distance runners in the marine realm: New insights into genetic diversity, kin relationships and social fidelity of Indian Ocean male sperm whales. Front Mar Sci 9:815684

Goudet J (2005) HEIRFSTAT, a package for R to compute and test hierarchical *F* -statistics. Mol Ecol Notes 5:184–186

Grosberg R, Cunningham CW (2001) Genetic structure in the sea. Marine community ecology. Sinauer Associates, Sunderland, MA, pp 61–84

Guichoux E, Lagache L, Wagner S, Chaumeil P, Léger P, Lepais O et al. (2011) Current trends in microsatellite genotyping. Mol Ecol Resour 11:591–611

Hanghøj K, Moltke I, Andersen PA, Manica A, Korneliussen TS (2019) Fast and accurate relatedness estimation from high-throughput sequencing data in the presence of inbreeding. GigaScience 8: giz034

Hayes SA, Josephson E, Maze-Foley K, Rosel PE, McCordic J, Brossard A et al. (2024) U.S. Atlantic and Gulf of Mexico Marine Mammal Stock Assessments 2023. NOAA Tech Memo NMFS-NE-321, Woods Hole, MA

Hemstrom W, Jones M (2023) snpR: User friendly population genomics for SNP data sets with categorical metadata. Mol Ecol Resour 23:962–973

Hodel RGJ, Segovia-Salcedo MC, Landis JB, Crowl AA, Sun M, Liu X et al. (2016) The report of my death was an exaggeration: a review for researchers using microsatellites in the 21st Century. Appl Plant Sci 4:1600025

Jacquard A (1972) Genetic information given by a relative. Biometrics 28:1101–1114

Jacquard A (1974) The genetic structure of populations. Springer-Verlag, Berlin

Jeffries DL, Copp GH, Lawson Handley L, Olsén KH, Sayer CD, Hänfling B (2016) Comparing RADseq and microsatellites to infer complex phylogeographic patterns, an empirical perspective in the Crucian carp, *Carassius carassius*, L. Mol Ecol 25:2997–3018

Jochens AD, Benoit-Bird K, Engelhaupt, D, Gordon J, Hu C, Jaquet N et al. (2008) Sperm whale seismic study in the Gulf of Mexico: Synthesis report. U.S. Dept. of the Interior, Minerals Management Service, Gulf of Mexico OCS Region: New Orleans, LA

Jombart T (2008) adegenet: a R package for the multivariate analysis of genetic markers. Bioinformatics 24:1403–1405

Kamvar ZN, Tabima JF, Grünwald NJ (2014) Poppr: an R package for genetic analysis of populations with clonal, partially clonal, and/or sexual reproduction. PeerJ 2:e281

Kelly RP, Palumbi SR (2010) Genetic structure among 50 species of the Northeastern Pacific rocky intertidal community. PLoS ONE 5:e8594

Kocher TD, Thomas WK, Meyer A, Edwards SV, Pääbo S, Villablanca FX et al. (1989) Dynamics of mitochondrial DNA evolution in animals: amplification and sequencing with conserved primers. Proc Natl Acad Sci USA 86:6196–6200

Konrad CM, Frasier TR, Whitehead H, Gero S (2019) Kin selection and allocare in sperm whales. Behav Ecol 30:194–201

Konrad CM, Gero S, Frasier T, Whitehead H (2018) Kinship influences sperm whale social organization within, but generally not among, social units. R Soc Open Sci 5:180914

Korunes KL, Samuk K (2021) PIXY: unbiased estimation of nucleotide diversity and divergence in the presence of missing data. Mol Ecol Resour 21:1359–1368

Lah L, Trense D, Benke H, Berggren P, Gunnlaugsson Þ, Lockyer C et al. (2016) Spatially explicit analysis of genome-wide SNPs detects subtle population structure in a mobile marine mammal, the harbor porpoise. PLoS ONE 11: e0162792

Lande R (1995) Mutation and Conservation. Conserv Biol 9:782–791

Leigh JW, Bryant D, Nakagawa S (2015) POPART: full-feature software for haplotype network construction. Methods Ecol Evol 6:1110–1116

Lemopoulos A, Prokkola JM, Uusi-Heikkilä S, Vasemägi A, Huusko A, Hyvärinen P et al. (2019) Comparing RADseq and microsatellites for estimating genetic diversity and relatedness — Implications for brown trout conservation. Ecol Evol 9:2106–2120

Lizewski K, Alexander A, Steel D, Mate B, Engelhaupt D, Wise J et al. (2025) Change in matrilineal structure over time in an isolated population of sperm whales. Mar Ecol Prog Ser 758:161–178

Lyrholm T, Leimar O, Johanneson B, Gyllensten U (1999) Sex-biased dispersal in sperm whales: contrasting mitochondrial and nuclear genetic structure of global populations. Proc R Soc B Biol Sci 266:347–354

Manichaikul A, Mychaleckyj JC, Rich SS, Daly K, Sale M, Chen W-M (2010) Robust relationship inference in genome-wide association studies. Bioinformatics 26:2867–2873

McCartney-Melstad E, Vu JK, Shaffer HB (2018) Genomic data recover previously undetectable fragmentation effects in an endangered amphibian. Mol Ecol 27:4430–4443

McKinney GJ, Waples RK, Seeb LW, Seeb JE (2017) Paralogs are revealed by proportion of heterozygotes and deviations in read ratios in genotyping-by-sequencing data from natural populations. Mol Ecol Resour 17:656–669

Mesnick SL, Taylor BL, Archer FI, Martien KK, Treviño SE, Hancock-Hanser BL et al. (2011a) Sperm whale population structure in the eastern and central North Pacific inferred by the use of single-nucleotide polymorphisms, microsatellites and mitochondrial DNA. Mol Ecol Resour 11:278–298

Mesnick SL, Taylor BL, Archer FI, Martien KK, Treviño SE, Hancock-Hanser BL et al. (2011b) Sperm whale population structure in the eastern and central North Pacific inferred by the use of single-nucleotide polymorphisms, microsatellites and mitochondrial DNA. Mol Ecol Resour 11:278–298

Mijangos JL, Gruber B, Berry O, Pacioni C, Georges A (2022) dartR v2: An accessible genetic analysis platform for conservation, ecology and agriculture. Methods Ecol Evol 13:2150–2158

Möller LM, Beheregaray LB (2004) Genetic evidence for sex-biased dispersal in resident bottlenose dolphins (*Tursiops aduncus*). Mol Ecol 13:1607–1612

Montana L, Bringloe TT, Bourret A, Sauvé C, Mosnier A, Ferguson SH et al. (2024) Reduced representation and whole-genome sequencing approaches highlight beluga whale populations associated to eastern Canada summer aggregations. Evol Appl 17: e70058

Morin PA, Foote AD, Baker CS, Hancock-Hanser BL, Kaschner K, Mate BR et al. (2018) Demography or selection on linked cultural traits or genes? Investigating the driver of low mtDNA diversity in the sperm whale using complementary mitochondrial and nuclear genome analyses. Mol Ecol 27:2604–2619

Nazareno AG, Bemmels JB, Dick CW, Lohmann LG (2017) Minimum sample sizes for population genomics: an empirical study from an Amazonian plant species. Mol Ecol Resour 17:1136–1147

O'Corry-Crowe G, Suydam R, Quakenbush L, Potgieter B, Harwood L, Litovka D et al. (2018) Migratory culture, population structure and stock identity in North Pacific beluga whales (Delphinapterus leucas). PLoS ONE 13:e0194201

Ortega-Ortiz JG, Engelhaupt D, Winsor M, Mate BR, Rus Hoelzel A (2012) Kinship of long-term associates in the highly social sperm whale. Mol Ecol 21:732–744

Palumbi SR (1994) Genetic divergence, reproductive isolation, and marine speciation. Annu Rev Ecol Syst 25:547–572

Palumbi SR (2003) Population genetics, demographic connectivity, and the design of marine reserves. Ecol Appl 13:146–158.

Pante E, Simon-Bouhet B (2013) marmap: A Package for importing, plotting and analyzing bathymetric and topographic data in R. PLOS ONE 8: e73051.

Paradis E (2010) pegas: an R package for population genetics with an integrated–modular approach. Bioinformatics 26:419–420

Petit-Marty N, Vázquez-Luis M, Hendriks IE (2021) Use of the nucleotide diversity in COI mitochondrial gene as an early diagnostic of conservation status of animal species. Conserv Lett 14:e12756

Pilot M, Dahlheim ME, Hoelzel AR (2010) Social cohesion among kin, gene flow without dispersal and the evolution of population genetic structure in the killer whale (*Orcinus orca*). J Evol Biol 23:20–31

Pinela AM, Quérouil S, Magalhães S, Silva MA, Prieto R, Matos JA et al. (2009) Population genetics and social organization of the sperm whale (*Physeter macrocephalus*) in the Azores inferred by microsatellite analyses. Can J Zool 87:802–813

Poelstra JW, Montero BK, Lüdemann J, Yang Z, Rakotondranary SJ, Hohenlohe P et al. (2022) RADseq data reveal a lack of admixture in a mouse lemur contact zone contrary to previous microsatellite results. Proc R Soc B Biol Sci 289:20220596

R Core Team (2024) R: a language and environment for statistical computing, R foundation for statistical computing, https://www.R-project.org/ (Vienna, Austria)

Rambaut A (2002) Se-Al v.2.0a11. http://tree.bio.ed.ac.uk/software/seal/

Rendell LE, Whitehead H (2003) Vocal clans in sperm whales (*Physeter macrocephalus*). Proc R Soc Lond B Biol Sci 270:225–231

Richard KR, Dillon MC, Whitehead H, Wright JM (1996) Patterns of kinship in groups of free-living sperm whales (*Physeter macrocephalus*) revealed by multiple molecular genetic analyses. Proc Natl Acad Sci 93:8792–8795

Rigby RA, Stasinopoulos DM (2005) Generalized additive models for location, scale and shape. J R Stat Soc Ser C Appl Stat 54:507–554

Rochette NC, Rivera-Colón AG, Catchen JM (2019) Stacks 2: analytical methods for paired-end sequencing improve RADseq-based population genomics. Mol Ecol 28:4737–4754

Rosel PE (2003) PCR-based sex determination in Odontocete cetaceans. Conserv Genet 4:647–649

Rosel PE, Hansen L, Hohn AA (2009) Restricted dispersal in a continuously distributed marine species: common bottlenose dolphins *Tursiops truncatus* in coastal waters of the western North Atlantic. Mol Ecol 18:5030–5045

Russello MA, Waterhouse MD, Etter PD, Johnson EA (2015) From promise to practice: pairing non-invasive sampling with genomics in conservation. PeerJ 3:e1106

Sanford E, Kelly MW (2011) Local adaptation in marine invertebrates. Annu Rev Mar Sci 3:509–535

Santiago E, Caballero A, Köpke C, Novo I (2024) Estimation of the contemporary effective population size from SNP data while accounting for mating structure. Mol Ecol Resour 24:e13890

Sarano F, Girardet J, Sarano V, Vitry H, Preud'homme A, Heuzey R et al. (2021) Kin relationships in cultural species of the marine realm: case study of a matrilineal social group of sperm whales off Mauritius island, Indian Ocean. R Soc Open Sci 8:201794

Sellas AB, Wells RS, Rosel PE (2005) Mitochondrial and nuclear DNA analyses reveal fine scale geographic structure in bottlenose dolphins (*Tursiops truncatus*) in the Gulf of Mexico. Conserv Genet 6:715–728

Sinclair C, Sinclair J, Zolman E, Martinez A, Balmer B, Barry K (2015) Remote biopsy sampling field procedures for cetaceans used during the natural resource damage assessment of the MSC252 Deepwater Horizon oil spill. NOAA technical memorandum, National Marine Fisheries Service, Southeast Fisheries Science Center, Pascagoula, MS

Solsona-Berga A, Frasier K, Posdaljian N, Baumann-Pickering S, Wiggins S, Soldevilla M et al. (2024) Accounting for sperm whale population demographics in density estimation using passive acoustic monitoring. Mar Ecol Prog Ser 746:121–140

Speakman TR, Wilcox LA, Balmer BC, Barry KP, Paterson C, Quigley BM et al. (2022) Fine-scale social and genetic structure of common bottlenose dolphins (*Tursiops truncatus*) in the Barataria Basin, Louisiana, USA. Aquat Conserv Mar Freshw Ecosyst 32:1437–1456

Templeton AR, Crandall KA, Sing CF (1992) A cladistic analysis of phenotypic associations with haplotypes inferred from restriction endonuclease mapping and DNA sequence data. III. Cladogram estimation. Genetics 132:619–633

Thia JA (2023) Guidelines for standardizing the application of discriminant analysis of principal components to genotype data. Mol Ecol Resour 23:523–538

Vasimuddin MD, Misra S, Li H, Aluru S (2019) Efficient Architecture-Aware Acceleration of BWA-MEM for MulticoreSystems. IEEE Parallel and Distributed Processing Symposium (IPDPS), https://doi.org/10.1109/IPDPS.2019.00041

Violi B, de, Jong MJ, Frantzis A, Alexiadou P, Tardy C, Ody D et al. (2023) Genomics reveals the role of admixture in the evolution of structure among sperm whale populations within the Mediterranean Sea. Mol Ecol 32:2715–2731

Vollmer NL, Rosel PE, Mullin KD, Schwacke LH, Garrison LP, Balmer BC et al. (2021) Assessing common bottlenose dolphin (*Tursiops truncatus*) population structure in Mississippi Sound and coastal waters of the north central Gulf of Mexico. Aquat Conserv Mar Freshw Ecosyst 31:2951–2966

Vollmer NL, Viricel A, Wilcox L, Katherine Moore M, Rosel PE (2011) The occurrence of mtDNA heteroplasmy in multiple cetacean species. Curr Genet 57:115–131

Waples RK, Albrechtsen A, Moltke I (2019) Allele frequency-free inference of close familial relationships from genotypes or low-depth sequencing data. Mol Ecol 28:35–48

Waples RS, Anderson EC (2017) Purging putative siblings from population genetic data sets: a cautionary view. Mol Ecol 26:1211–1224

Weir BS, Cockerham CC (1984) Estimating F-statistics for the analysis of population structure. Evolution 38:1358–1370

Whitehead H (1993) The behaviour of mature male sperm whales on the Galápagos Islands breeding grounds. Can J Zool 71:689–699

Whitehead H (1998) Cultural selection and genetic diversity in matrilineal whales. Science 282:1708–1711

Whitehead H (2018) Sperm whale. In: Encyclopedia of Marine Mammals. Elsevier, Cambridge, MA, p 919–925

Whitehead H, Shin M (2022) Current global population size, post-whaling trend and historical trajectory of sperm whales. Sci Rep 12:19468

Whitehead H, Barlow J, Dawson SM, Eguiguren A, Miller B Shabangu F (2025) Physeter macrocephalus. The IUCN RedList of Threatened Species 2025:e.T41755A50382216. https://doi.org/10.2305/IUCN.UK.2025-2.RLTS.T41755A50382216.en. Accessed on 03 February 2026

Zimmerman SJ, Aldridge CL, Oyler-McCance SJ (2020) An empirical comparison of population genetic analyses using microsatellite and SNP data for a species of conservation concern. BMC Genomics 21:382

ACKNOWLEDGEMENTS

We thank Nicole Phillips for assistance with processing samples in the laboratory and Laura Dias, Jesse Wicker, and Wayne Hoggard for contributing to sample collection. Funding for sample collection was provided by the Bureau of Ocean Energy Management IA M09PG00014 and M11PG00041. This work made use of the Sedna high-performance computing cluster hosted at and administered within NOAA Fisheries, which has been funded by NOAA Fisheries' advanced technologies strategic initiatives. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the author(s) and do not necessarily reflect those of NOAA or the Department of Commerce.

AUTHOR CONTRIBUTIONS

RSB conducted the analysis, made figures, and wrote the manuscript. LAWT conducted lab work and analyzed the mitochondrial data, while LAWT, NLV, and PER conceptualized the project and helped to interpret results. AM, LPG, and DE collected samples. All authors helped to revise the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

RESEARCH ETHICS STATEMENT

This research was carried out, in part, under the auspices of the Cooperative Institute for Marine and Atmospheric Studies (CIMAS), a Cooperative Institute of the University of Miami and the National Oceanic and Atmospheric Administration, cooperative agreement #NA20OAR4320472. Biopsy sampling was conducted under NMFS MMPA permits 738, 779-1339, 779-1633, and 14450.

ADDITIONAL INFORMATION

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41437-026-00824-x.

Correspondence and requests for materials should be addressed to Reid S. Brennan.

Reprints and permission information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.